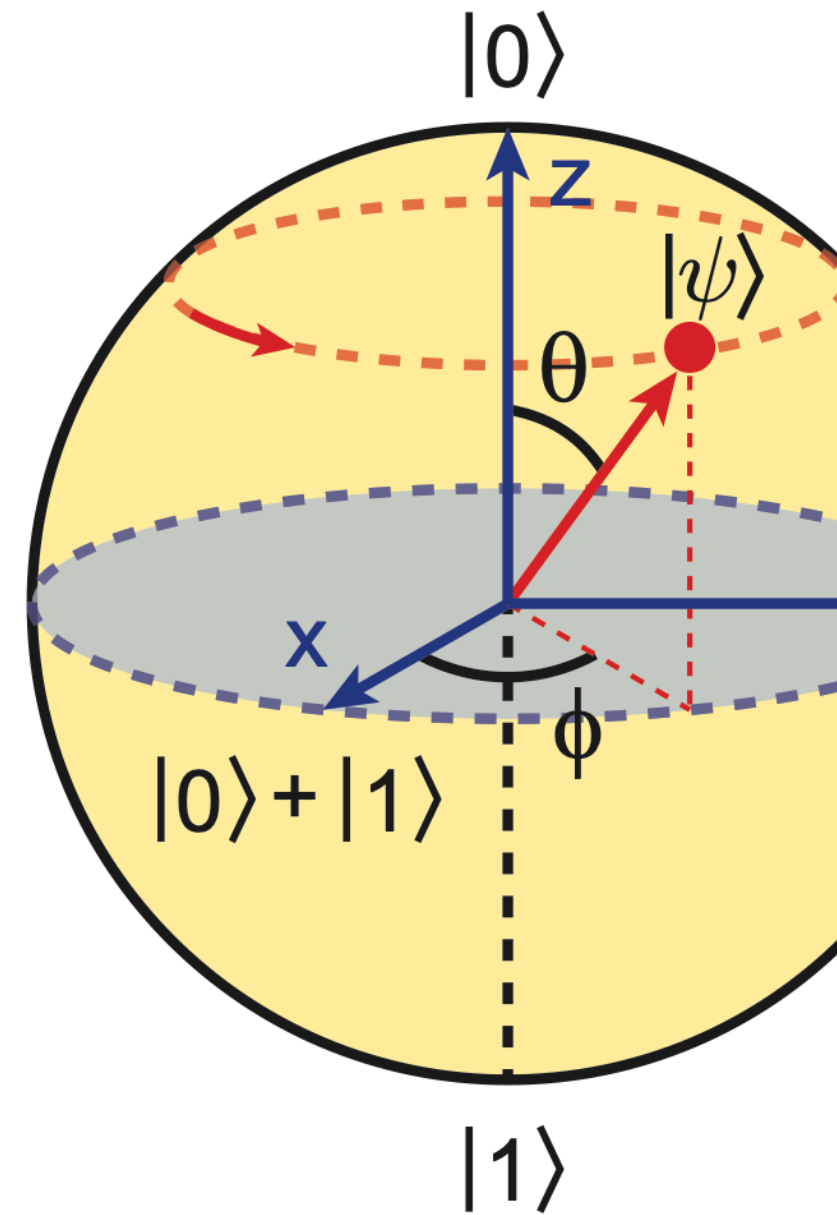


# Single-qubit gate infidelity $10^{-4}$ in a superconducting qubit

Zhiyuan Li et al.  
arXiv:2302.08690 (2023)



# Abstract

In this work:

1. Transmon qubit with long coherence time and fast, non-adiabatic control of single-qubit with
  - Average gate error below  $10^{-4}$  by randomized benchmarking.
2. Error analysis:
  - An error budget of  $(4.62 \pm 0.04) \times 10^{-5}$  for decoherence (lower bound).
  - A leakage rate per gate  $(1.16 \pm 0.04) \times 10^{-5}$ .
3. Obtain process matrix using GST.
4. Observation of non-Markovian behavior in long sequence GST.

# Experiment 1: Calibration of $\delta f$ , $\Omega_0$

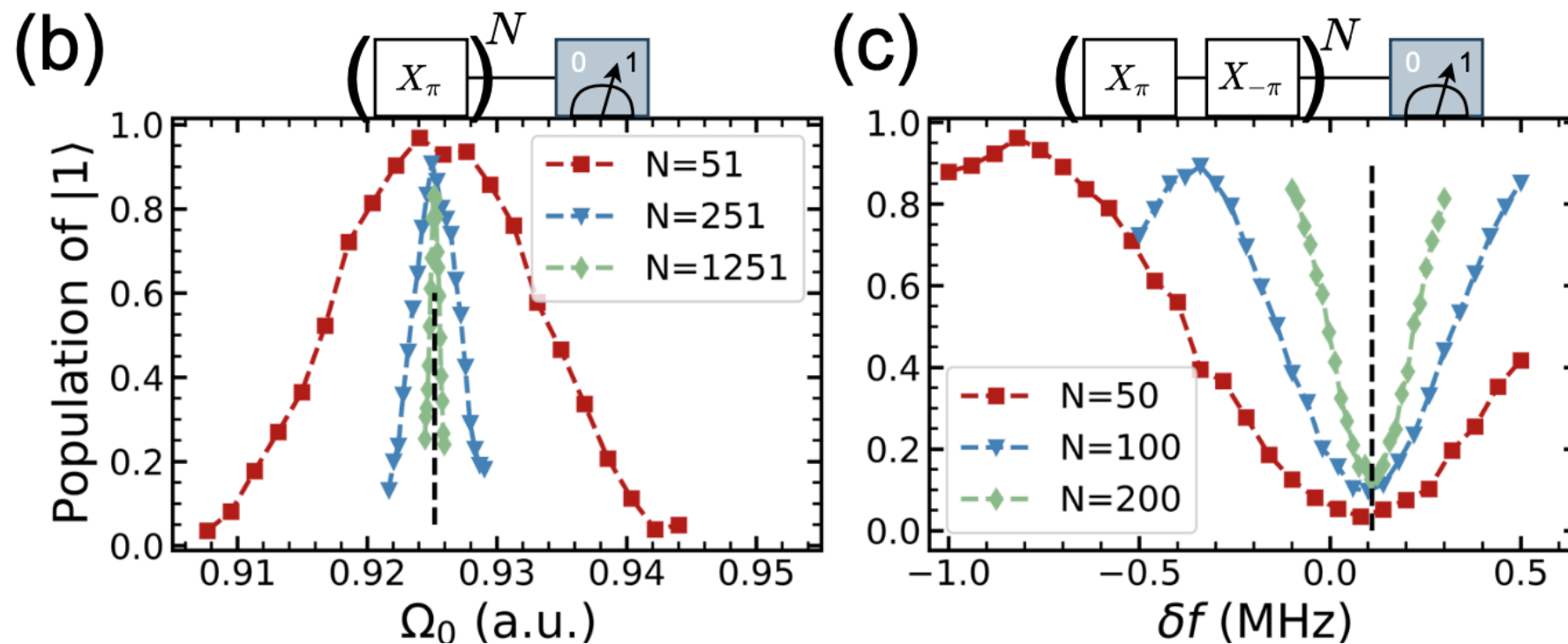
Microwave pulse with cosine-shaped envelope ( $t_g = 20$  ns)

$$\Omega(t) = \Omega_0(1 - \cos(2\pi t/t_g))$$

Suppress leakage and phase errors using DRAG scheme

$$\Omega_{DRAG}(t) = e^{i2\pi\delta f t} (\Omega(t) - i\alpha\dot{\Omega}(t)/\Delta)$$

Both  $\delta f$  and  $\alpha$  was used (which leads me to the Zijun Chen 2015 paper, which is an excellent article).

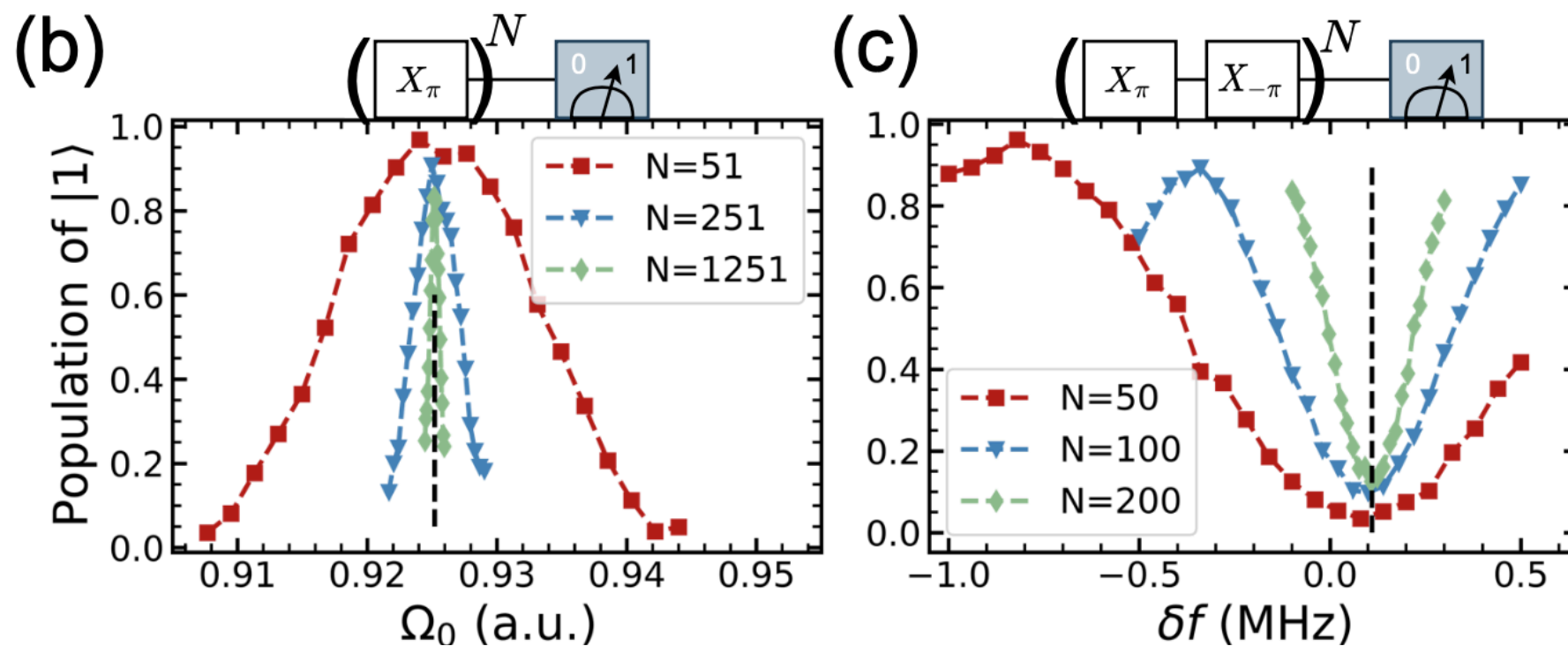


# Experiment 1: Calibration of $\delta f, \Omega_0$

Key takeaways:

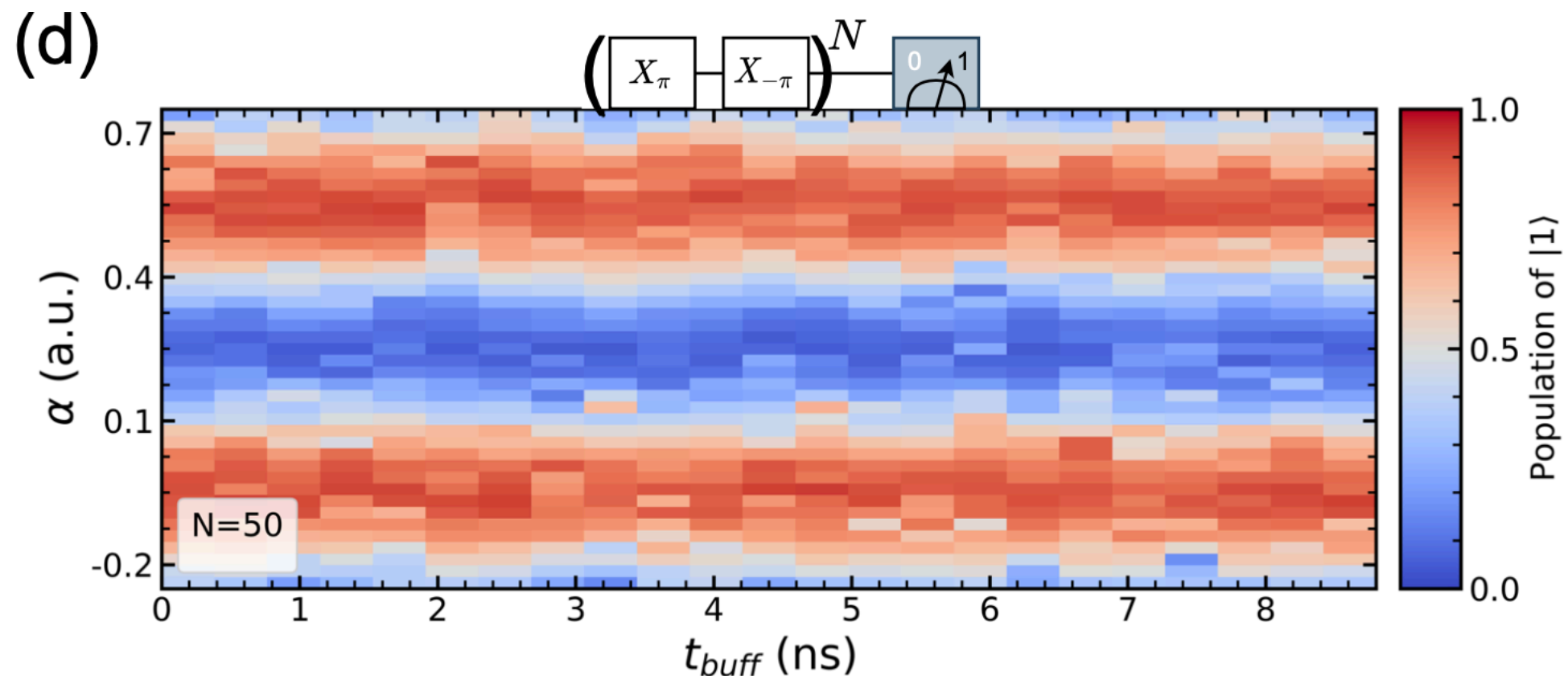
- Coherent errors is amplified by longer gate sequences (Sheldon et al. 2015)  $\epsilon_{coherent} \propto N$ .
- Increasing sequence length also improves precision of the metrology.

It's interesting to test if  $\Omega_0$  is the same for Sheldon 2015's protocol and Li 2023's protocol. In Sheldon scheme what we are measuring is how far the final state deviates from the original 1/2 pulse.



# Experiment 2: Calibration of buffer time

Spurious signal overlap is also a problem to be taken care of. Buffer time is added. To shorten the buffer time, the trailing edge of the driving pulse is measured.

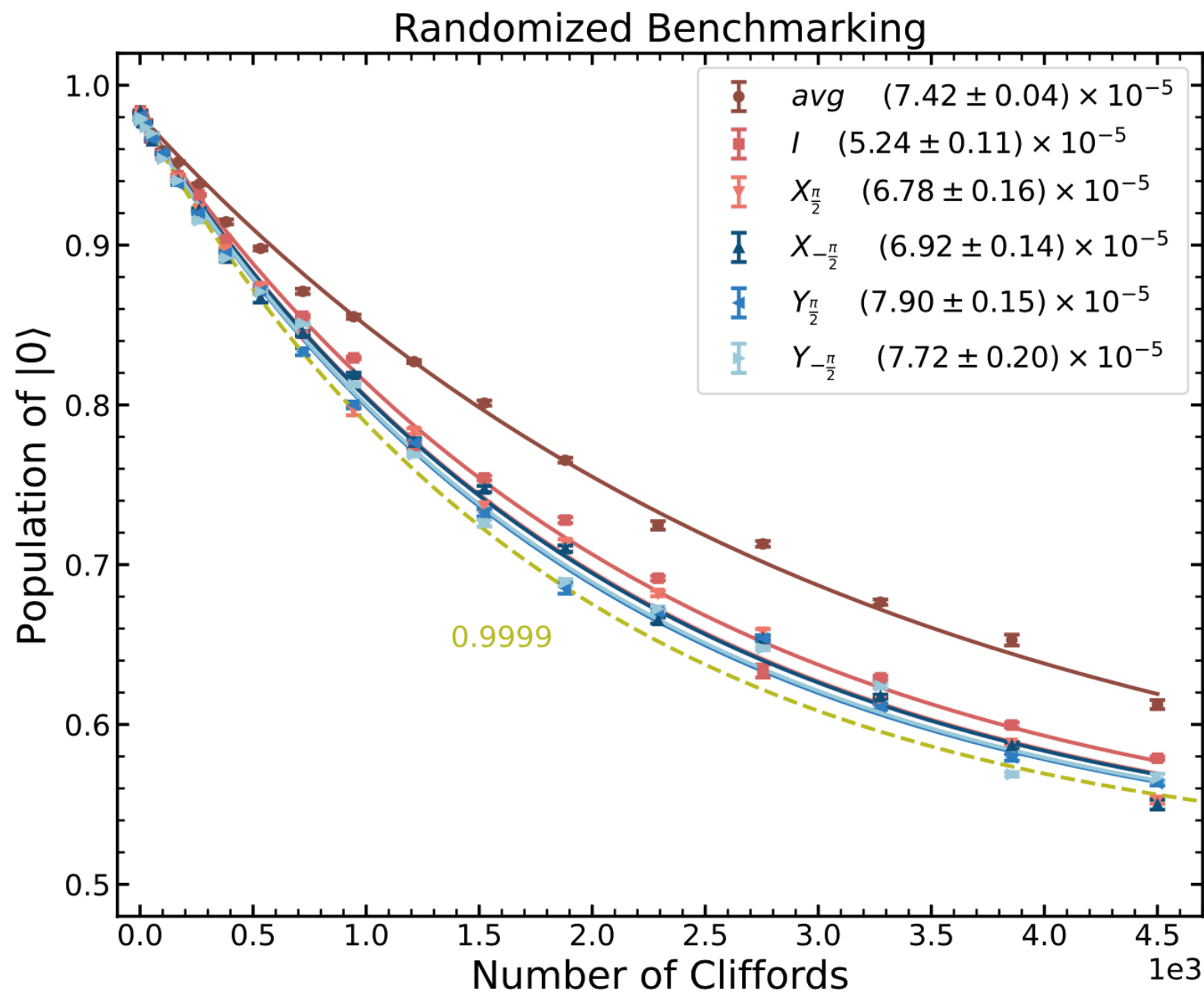


No visible movement is observed. They eventually added two nanoseconds into the total time to guarantee there is no overlap between pulses.

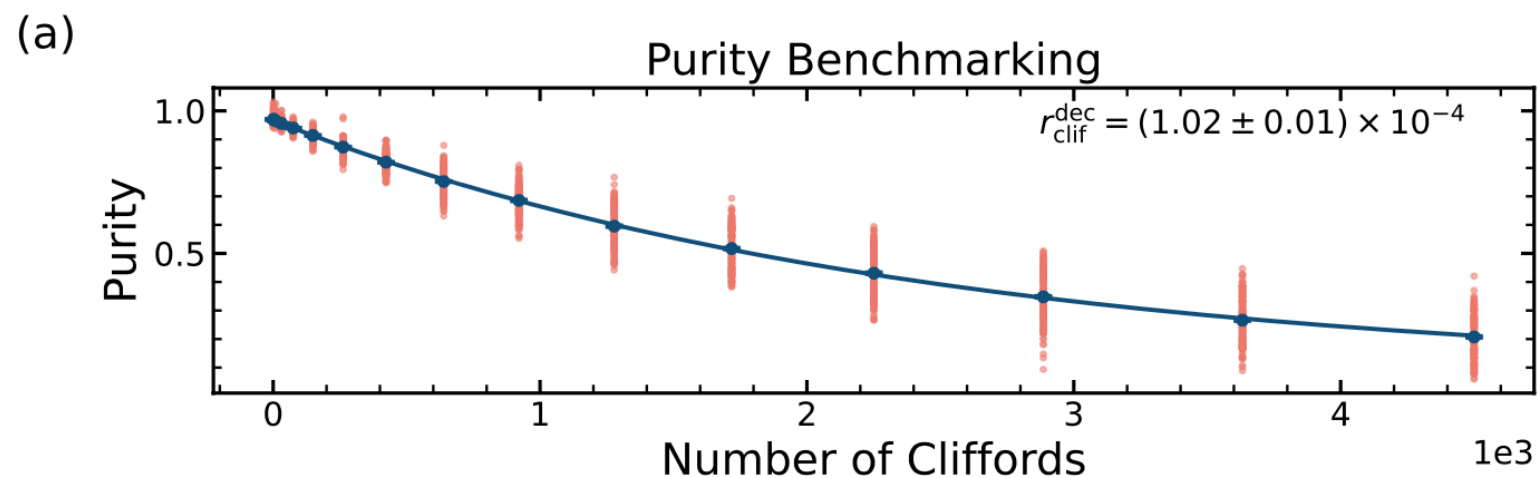
# Experiment 3: RB and interleaved RB

$$r_C = (1 - p_{avg})(1 - 1/2)$$

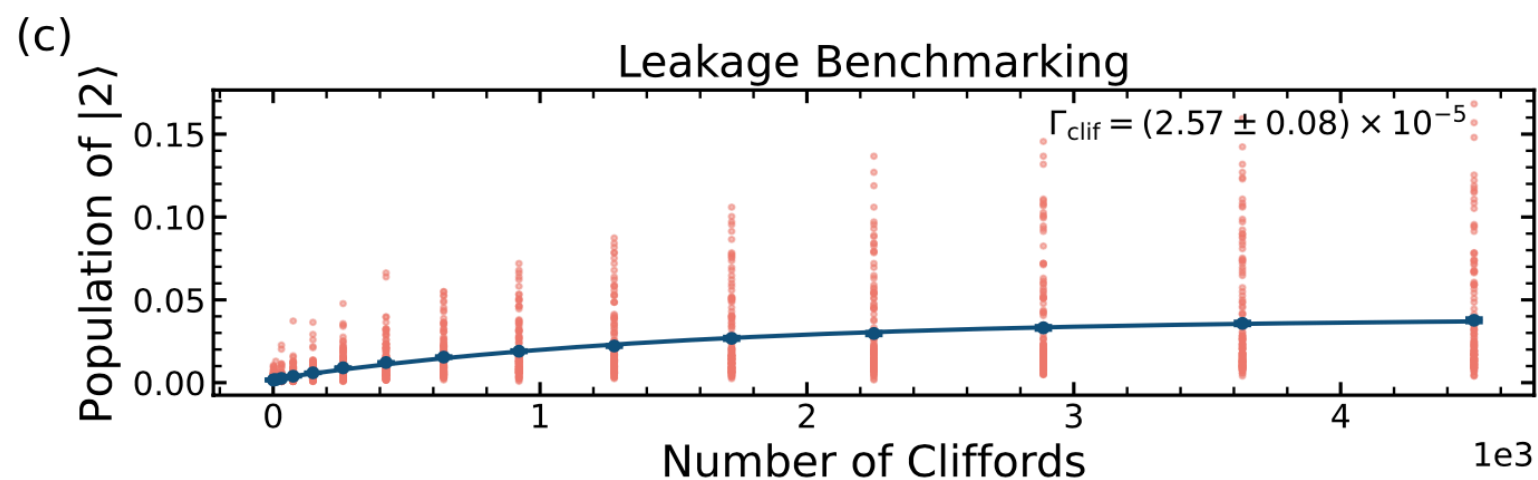
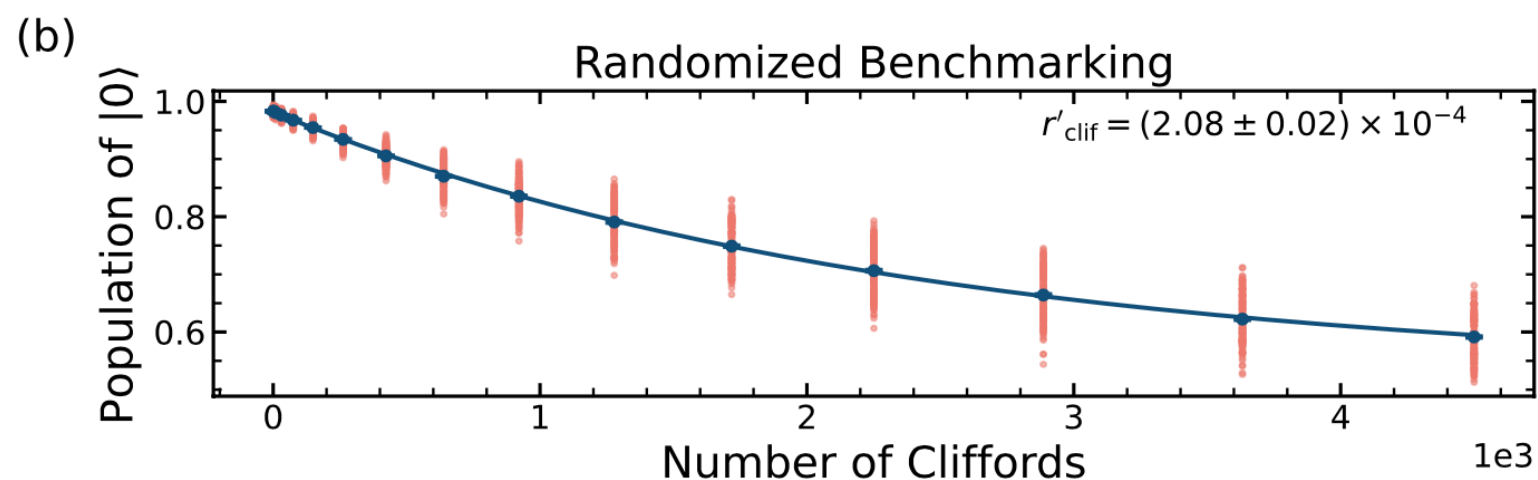
$$r_G = (1 - p_{int}/p_{avg})(1 - 1/2)$$



# Experiment 4: Purity benchmarking



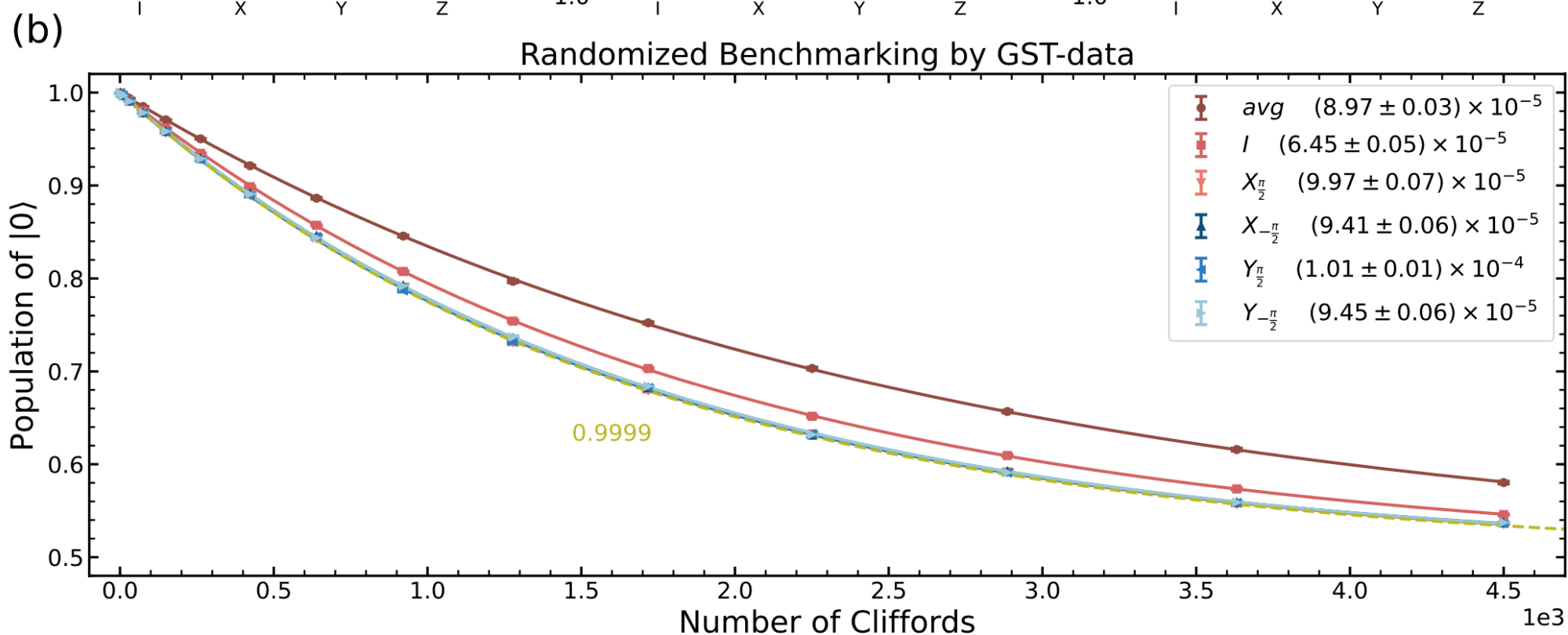
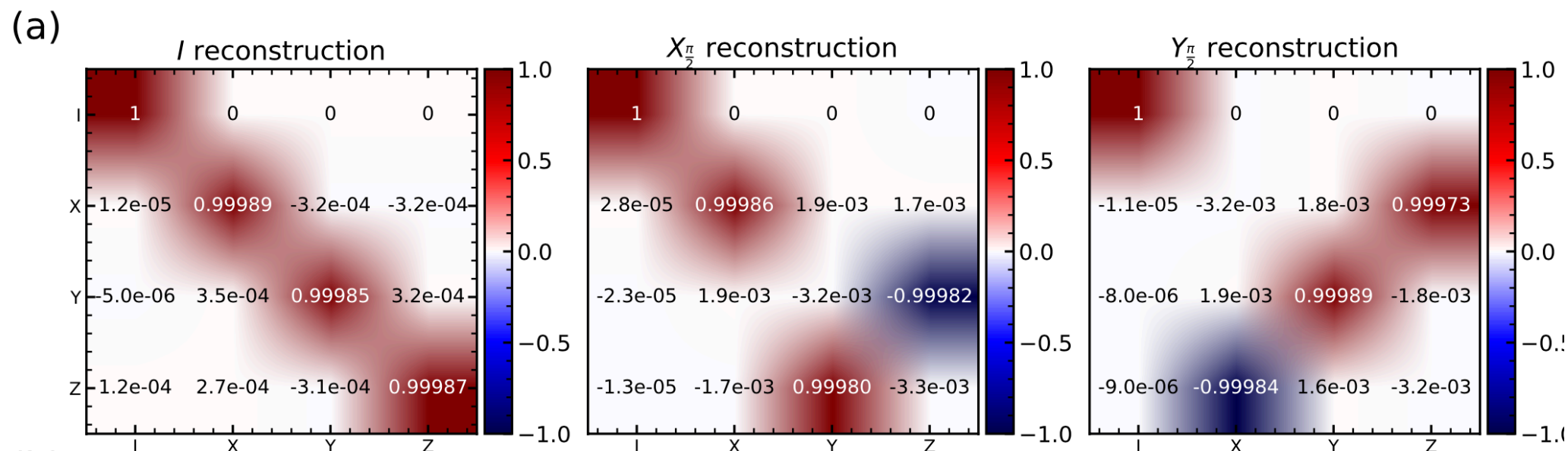
$$\text{Purity} = \langle \text{Tr}(\rho^2) \rangle_m$$



$$p_{2,m} = p_{2,\infty}(1 - e^{-\Gamma m}) + p_{2,0}e^{-\Gamma m}$$



# Experiment 5: Gate set tomography



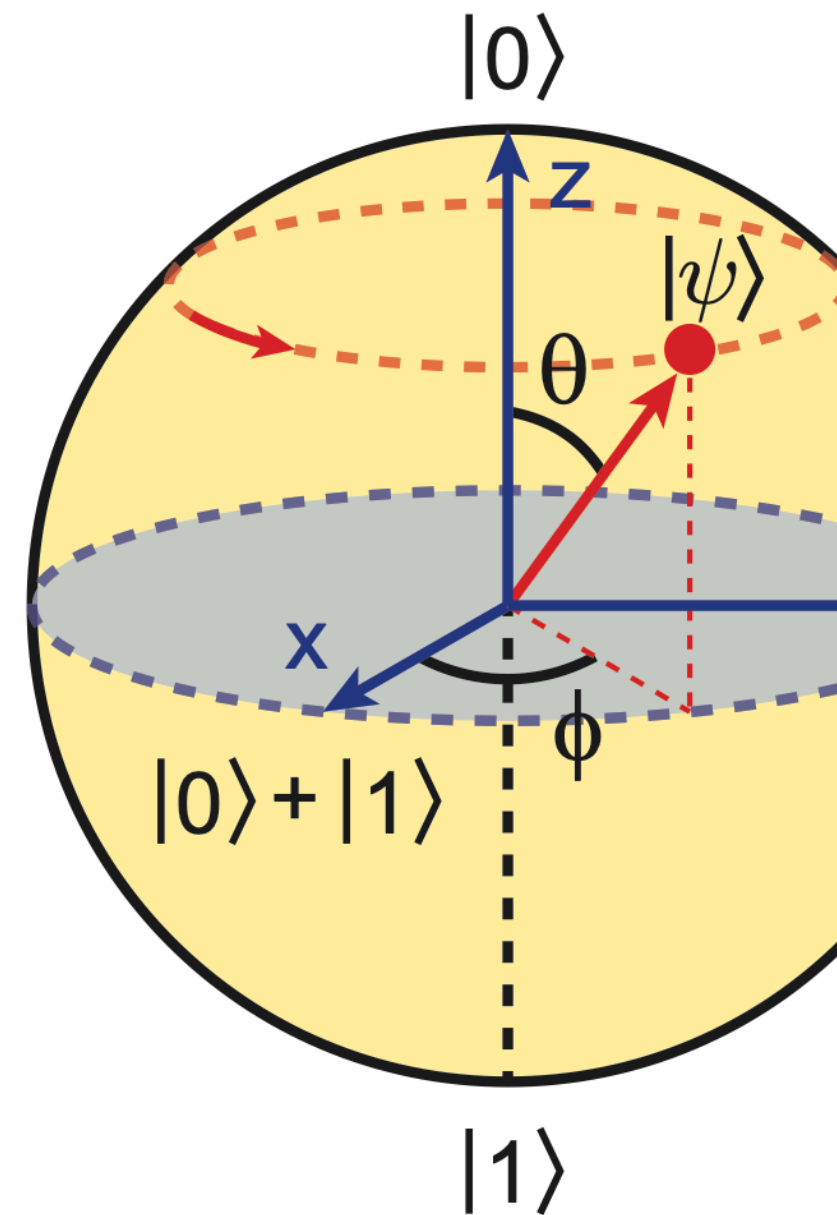
$$G \equiv \Lambda(\rho), \mathcal{P} = (I, X, Y, Z)$$

also some non-Markovianity profiles, but not convincing



# Measuring and suppressing quantum state leakage

Zijun Chen et al.  
Phys. Rev. Lett. 116, 020501 (2016)



# Abstract

In this work:

1. Quantification of leakage errors, using RB in conjunction with measurement of  $|2\rangle$ .
2. Characterization of single-qubit gate by refining use of DRAG along with detuning of pulses  $\rightarrow$  gate error below  $10^{-3}$  and leakage rates at  $10^{-5}$ . (basically decouple  $\alpha$  from correcting phase error).
3. Dependence of leakage on pulse length.
4. Remaining leakage is due to incoherent heating of the qubit.

Actually it wasn't until reading this paper that I had that “aha” moment. Need to remind that leakage out of qubit's manifolds process are unitary, coherent, but not a complete CPTP map (since Pauli matrices are  $2 \times 2$ ).

So, when discussing (Blok et al. PRL 2021) qutrit randomized benchmarking, we have to note that it's not the whole story.

And interestingly enough QEC is more vulnerable to leakage (Fowler PRA 2012).

# Experiment 1: Novel use of RB

Given a control envelope  $\Omega(t)$ , we add the time derivative to the quadrature component

$$\Omega_{DRAG} = \Omega(t) - i\alpha\dot{\Omega}(t)/D$$

This helps suppressing the spectral weight of the control pulse at the  $1 \leftrightarrow 2$  transition ( $\alpha = 1$  for leakage,  $\alpha = 0.5$  for phase errors),

$$W(\omega) = \left[1 + D(\omega - \omega_0)/\Delta\right] \int e^{-i\omega t} x(t) e^{i\omega_0 t} dt$$

Now, fix gate time ( $t_g = 10$  ns),  $x(t)$  being cosine-shaped, perform standard randomized benchmarking.

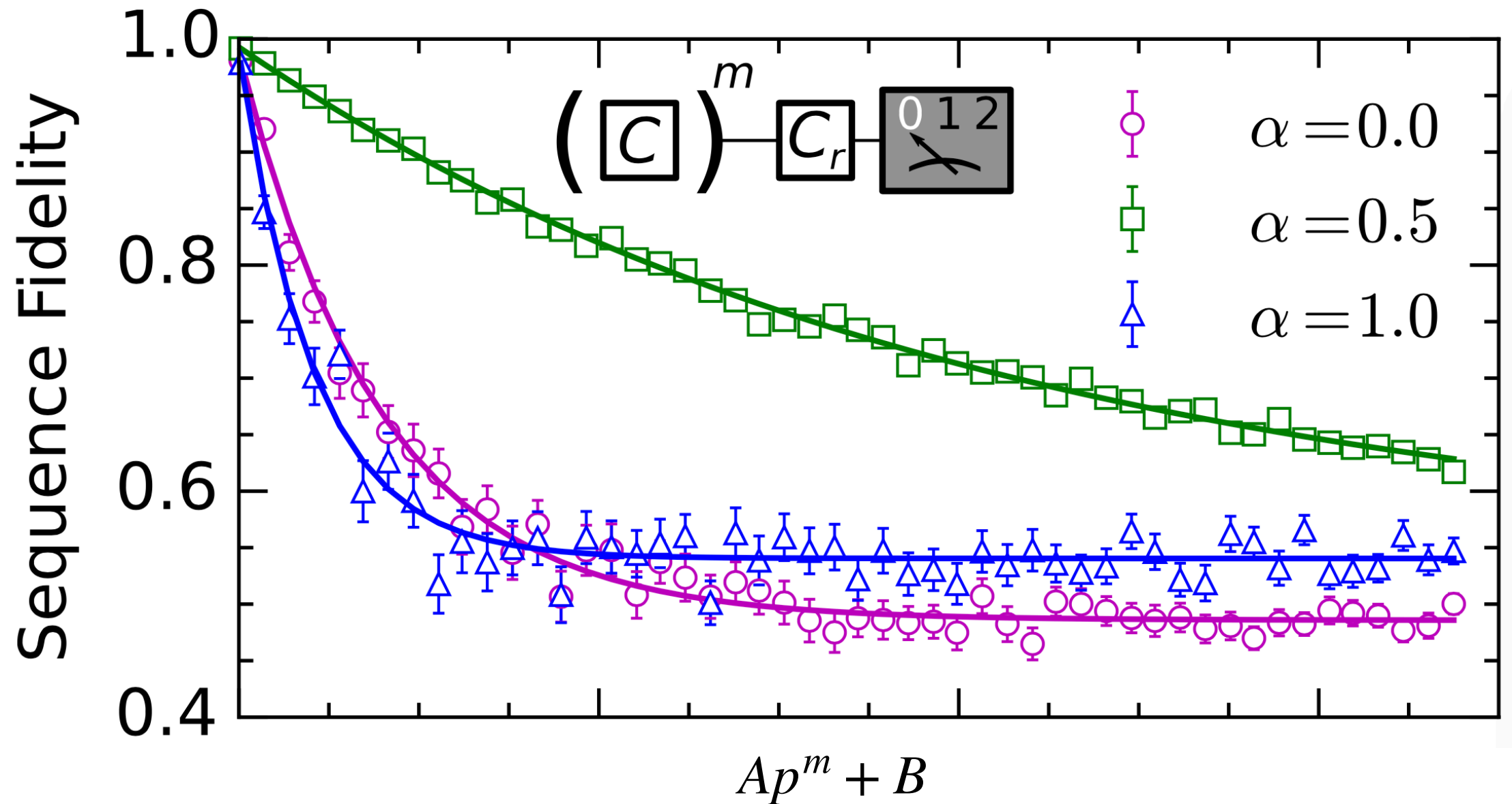
Results:

1.  $\epsilon_{avg} = 9.6 \pm 0.1 \times 10^{-4}$  ( $\alpha = 0.5$ )
2.  $\epsilon_{avg} = 6.3 \pm 0.2 \times 10^{-3}$  ( $\alpha = 0.0$ )
3.  $\epsilon_{avg} = 1.20 \pm 0.01 \times 10^{-2}$  ( $\alpha = 1.0$ )

Phase errors are  
more deleterious to  
RB metrics

# Experiment 1: Novel use of RB

$$\epsilon_{avg} = 9.6 \pm 0.1 \times 10^{-4} (\alpha = 0.5), 6.3 \pm 0.2 \times 10^{-3} (\alpha = 0.0), 1.20 \pm 0.01 \times 10^{-2} (\alpha = 1.0)$$



## Experiment 2: Novel use of RB (cont.)

In this experiment, leakage is characterized in the gateset, using RB simultaneously. For all  $\alpha$ , population of  $|2\rangle$  is monitored, which shows an exponential approach to saturation.

[*Comment:* For our case,  $|3\rangle$  has to be monitored. Previous experiments (February 2023) saw incredible difficulty as the shift of the cavity is no longer visible on the IQ plane with current use of measurement technique. This constitutes an experimental challenge]

To obtain leakage rate, fit to rate equation.

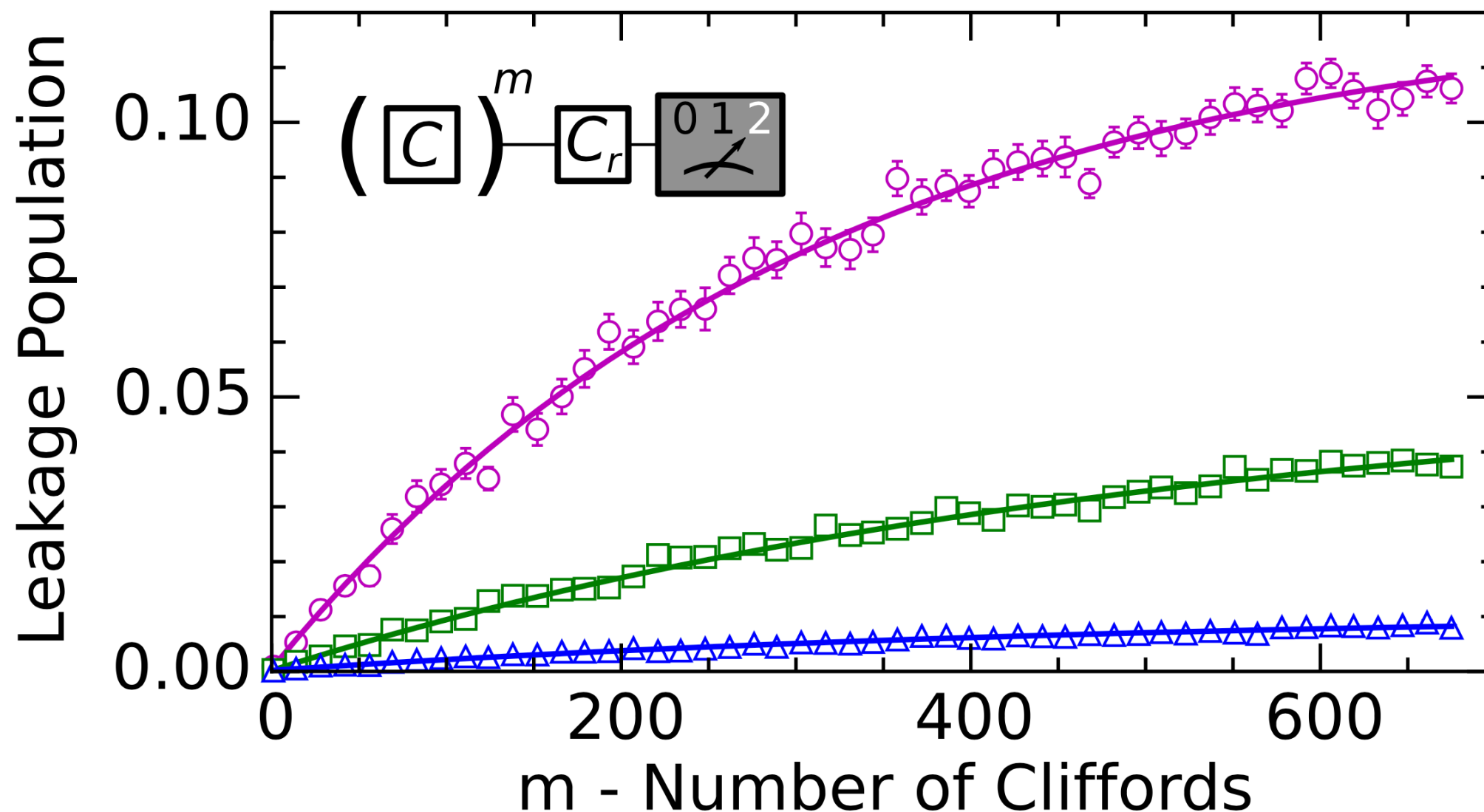
$$p_{2,m} = p_{2,\infty}(1 - e^{-\Gamma m}) + p_{2,0}e^{-\Gamma m}$$

Results:

1.  $\alpha = 0.0: 3.92 \pm 0.08 \times 10^{-4}$
2.  $\alpha = 0.5: 1.02 \pm 0.02 \times 10^{-4}$
3.  $\alpha = 1.0: 2.18 \pm 0.08 \times 10^{-5}$

## Experiment 2: Novel use of RB (cont.)

$$p_{2,m} = p_{2,\infty}(1 - e^{-\Gamma m}) + p_{2,0}e^{-\Gamma m}$$



→ Can't minimize both phase error and leakage error (with a single control's DOF)

# Experiment 3: Static detuning

To simultaneously optimize for both gate fidelity and leakage performance, we'd like to minimize leakage using simple DRAG, then separately compensate the AC Stark shift.

[*Comment:* In qutrit context, the phase errors doesn't necessarily come from level repulsion. Yes, Blok said something like this

$$A \cos \omega_{01} t (\hat{a} + \hat{a}^\dagger) \rightarrow H_{actual}^{ROT} = H_{ideal}^{ROT} + \text{some rotating terms off-diagonal,}$$

where the off-diagonal terms ( $\propto \Omega, t$ ) can be thought of as off-resonant drive of the  $(1 - 2)$  transitions. So in essence what is a *logical error* in qutrit community turns out to be a trivial, ubiquitous phase error in qubit community (Martinis PRA 2014). This is also in agreement with analysis of Fischer et al. 2022, where the phase acquired by the idling states are time-dependent (because the off-diagonal terms are time-dependent  $\propto e^{-i\alpha t}$ ).

However, this is not the only phase error we'll have to deal. Since charge dispersion is large for  $(1 - 2)$ , off-resonance from the qubit-subspace transition also introduces phase uncertainties, coupled by the  $\sigma_z$  contribution. These are not level-repulsion-origin phase errors]



# Experiment 3: Static detuning

To simultaneously optimize for both gate fidelity and leakage performance, we'd like to minimize leakage using simple DRAG, then separately compensate the AC Stark shift.

[*Comment:* In qutrit context, the phase errors doesn't necessarily come from level repulsion...]

To separately compensate AC Stark shift, we introduce a *static detuning* to the drive.

[*Comment:* In the 2009 paper Motzoi et al. shows that ramping the phase of pulses can improve the fidelity of the gate. But we don't have that luxury]

Turns out a static detuning works like a charm with the DRAG scheme,

$$\Omega_{DRAG}^{detuned} = \Omega_{DRAG} e^{2\pi i \delta f t},$$

where  $\delta f$  is the static detuning of the drive from the resonance frequency. The scheme involves a pseudo-identity operator  $X_{\pi} X_{-\pi}$  to sweep  $\delta f$ .

To verify if the phase error is suppressed, use GST.



# Experiment 3: Static detuning

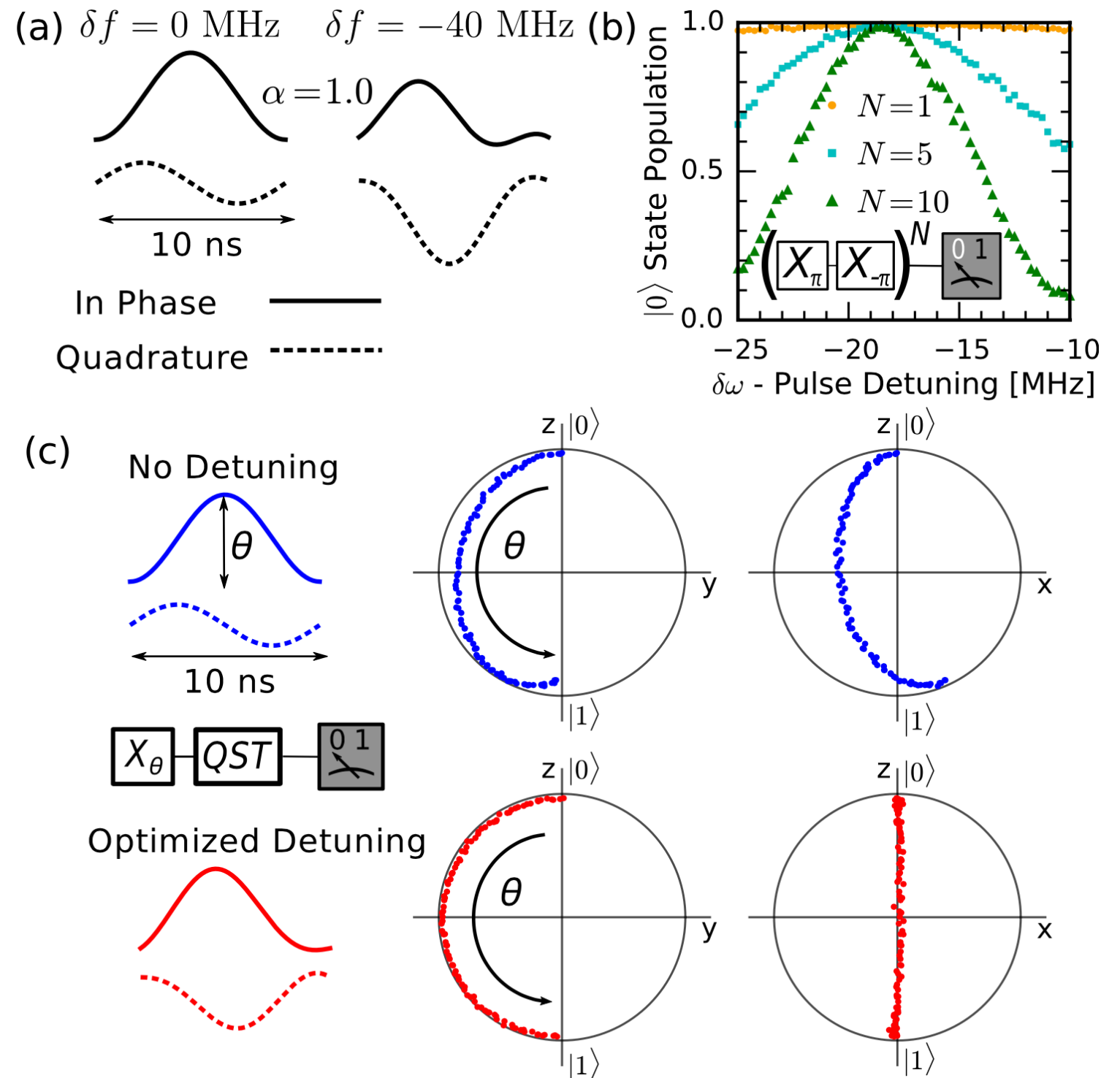
Original DRAG: Motzoi et al.  
Phys. Rev. Lett. 103, 110501 (2009)

Static detuning: Gambetta et al.  
Phys. Rev. A 83, 012308 (2011)  
see also Martinis paper 2014.

Increase of sequence length improves precision.

State trajectories on BS obtained using standard GST.

It's possible to outsource the task of reducing phase error to pulse detuning *which is mind-blowing*.



# Experiment 4: Decoupled $\alpha$ from phase error

We now explore more in detail the dependence of fidelity and leakage on  $\alpha$ .

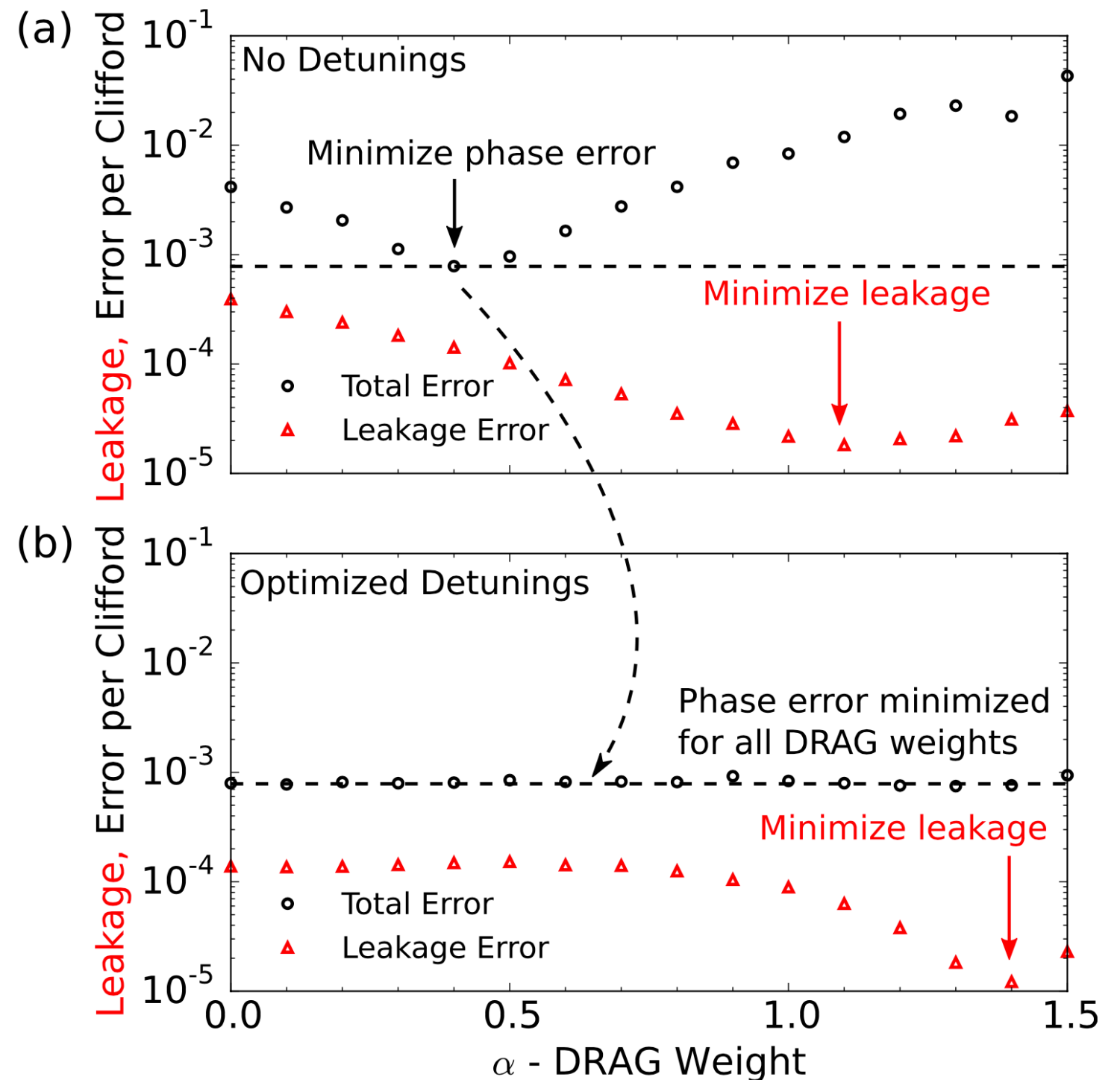
1. Without detuning, best  $\alpha = 0.4$  yields minimum error per Clifford below  $7.9 \pm 3 \times 10^{-4}$  (deviation from 0.5 probably due to pulse distortion) [Comment: We also observe both absolute magnitude and sign deviation. While absolute magnitude can be resourced to signal distortion, sign deviation can be understood as sign convention of DRAG on IBM devices]. Away from  $\alpha$  optimum errors increase rapidly.
2. *With* detuning, error per Clifford gate shows no dependence on  $\alpha$  (actually with some more calibrations, not just only  $\delta f$ ).
3. Now that  $\partial\epsilon_{\text{Clifford}}/\partial\alpha \rightarrow 0$ , leakage to  $|2\rangle$  is monitored to find best  $\alpha$ .
  - Without detuning, best  $\alpha = 1.1$ ,  $\Gamma \sim 1.82 \pm 0.07 \times 10^{-5}$
  - For  $\alpha > 0.4$ , since the pulse spectra overlaps more with  $\omega_{01} + \Delta$ , leakage increases.
  - However, increasing to  $\alpha = 1.4$ , leakage rate  $\Gamma$  is as low as  $10^{-5}$ .

# Experiment 4: Decoupled $\alpha$ from phase error

Takeaway: Both errors can be suppressed at the same time.

Error per Clifford  
 $8.7 \pm 0.4 \times 10^{-4}$   
Leakage rate  $\Gamma$   
 $1.2 \pm 0.1 \times 10^{-5}$

Zijun Chen et al.  
Phys. Rev. Lett. 116, 020501 (2016)



# Experiment 5: Leakage dependence on gate duration

As noted, pulse detuning can affect the leakage rate (trivial). For simplicity, we set  $\delta f = 0$  throughout the experiment.

1. Set  $\alpha = 0$  and varies  $t_g$  between 8 ns and 50 ns.
2. Set  $\alpha = 1.1$  and varies  $t_g$  between 8 ns and 50 ns.

Results: For short pulses, leakage rate decreases exponentially with increasing pulse length. DRAG supports that exponential reduction by an order of magnitude or more.

For longer pulses (i.e. past 15 ns), leakage rate levels off and tends to increase. DRAG effect is no longer distinguishable for pulses longer than 20 ns. This suggests leakage is due to heating or noise at  $\omega_{12}$  frequency, rather than coherent processes.

(Experiment 6: to measure incoherent leakage rate, basically measure  $T_1$  again.)

# Experiment 5: Leakage dependence on gate duration

Takeaway: DRAG matters only when gate are short (which is great!)

Zijun Chen et al.  
Phys. Rev. Lett. 116, 020501 (2016)

Why great? Because all we can do is confined in the parameter regime in which gate duration must be short so that Fourier broadening can compensate for large charge dispersion

